

# Jelin Raphael Akkara

## PhD Researcher in Embodied AI

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### RESEARCH PROFILE

Embodied AI researcher working on multimodal failure recovery, vision-language reasoning, and long-horizon planning. First-author paper accepted at IROS 2026, with coauthored publications at ICRA and IJCV.

### SELECTED PUBLICATIONS

- **Akkara, J. R.**, et al. IMPRINT: Image-Conditioned Query Enrichment for Long-Tail Object Goal Navigation. IEEE/RSJ **IROS 2026** (accepted). [arxiv.org/abs/2607.25106](https://arxiv.org/abs/2607.25106)
- Ziliotto, F., **Akkara, J. R.**, et al. PersONAL: Towards a Comprehensive Benchmark for Personalized Embodied Agents. IEEE **ICRA 2026**. [arxiv.org/abs/2509.19843](https://arxiv.org/abs/2509.19843)
- Ziliotto, F., **Akkara, J. R.**, et al. PersONAL-3D: A Comprehensive Benchmark for Personalized Embodied AI Agents. International Journal of Computer Vision (**IJCV**), 2026. [doi.org/10.1007/s11263-026-02922-z](https://doi.org/10.1007/s11263-026-02922-z)

### RESEARCH EXPERIENCE

- **PhD Researcher in Embodied AI** 2025 - Present  
University of Padova / Fondazione Bruno Kessler (FBK)  
Research on multimodal perception and decision-making for embodied agents, spanning object navigation, personalized interaction, and adaptation to novel failures in long-horizon tasks.
- **GenAI Research Intern** 2025  
ViWear  
Improved a Stable Diffusion-based virtual try-on model by 7–12% on perceptual metrics and increased inference throughput from 1 to 3 frames per second through fine-tuning and architectural changes.
- **Computer Vision Research Intern** 2024  
VIMP, University of Padova / Fondazione Bruno Kessler (FBK)  
Built a modular visual mapping framework with 2D semantic maps and multimodal embeddings; reduced map memory use by 99.75% relative to the dense mapping baseline.

### EDUCATION AND TRAINING

- **PhD in Brain, Mind and Computer Science** 2025 - Present  
University of Padova / Fondazione Bruno Kessler (FBK)
- **MSc in Physics of Data** 2022 - 2025  
University of Padova  
GPA: 110 / 110
- **B.Tech in Engineering Physics** 2018 - 2022  
National Institute of Technology, Calicut (NITC)  
First Class with Distinction

### RELEVANT SKILLS

- **Research:** Embodied AI, object goal navigation, computer vision, vision-language models, multimodal learning, generative modeling, simulation-based evaluation
- **Tools:** Python, PyTorch, TensorFlow/Keras, Habitat AI, AI2-THOR, Hugging Face, Docker, Linux, SLURM